puplis
EXAM SYSTEM

Exams that run even when the network doesn’t.

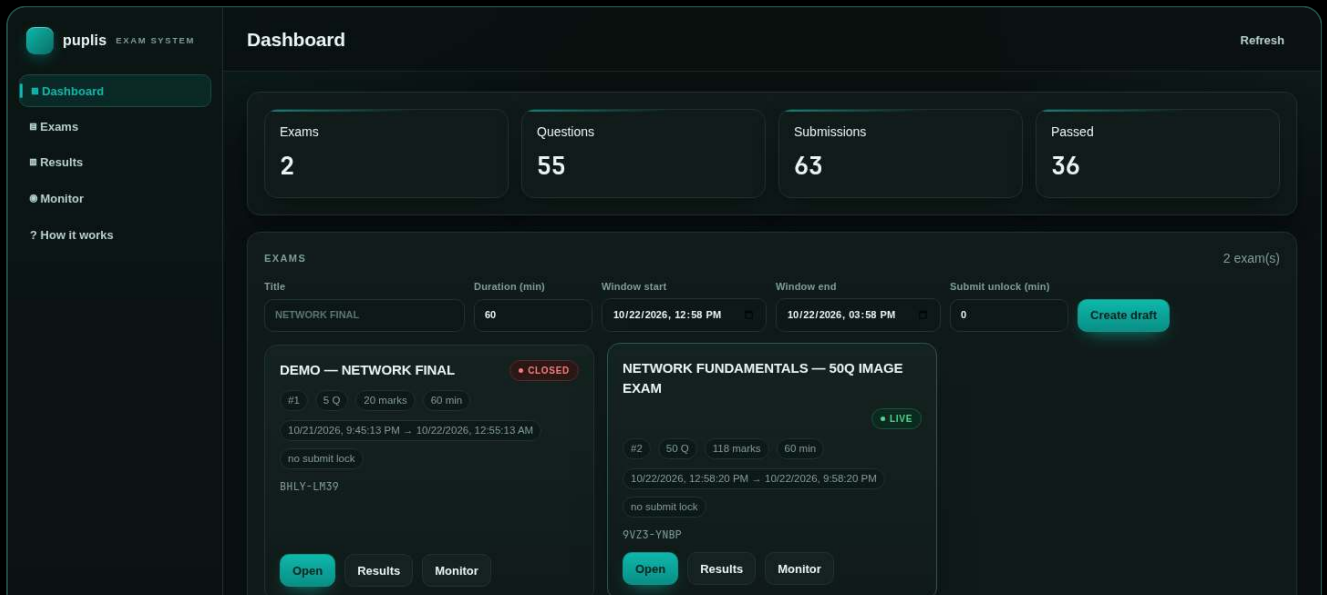
The exam platform in daily use at Kalar Technical Institute. The committee prepares the paper, students sit it in the browser on the lab LAN, and one consistent grading engine marks every answer. No app install. No internet needed during the exam. No lost submissions.

Admin web app · port 8013 | Student web app · port 8014 | Offline-capable | Self-hosted on one lab PC

57
exams sat in one academic year

5,700+
student results recorded

50
students start in 648 ms



PROVEN IN DAILY USE

Already running real exams. Every week of the semester.

puplis is not a prototype. It has completed a full academic year at Kalar Technical Institute: **57 exams sat**, **4,862 questions** delivered and **more than 5,700 student results** recorded across database, networks, programming, IT, mathematics, English and more. Teachers send the paper; the lab manager imports, publishes and the sitting happens the same day.

57	4,862	5,700+	1 year
exams delivered	questions in the bank	results marked	of continuous use

Every exam, one repeatable pipeline

01 · PREPARE
Build the paper
Question editors or import the committee's CSV format, validated row by row.

02 · SCHEDULE
Set the window
Duration, time window, shuffling, marking policy and optional submit lock.

03 · SIT
Students take it
Browser on the lab LAN. Shuffled paper, live timer, offline-safe.

04 · MARK
Automatic grading
Every question type marked by one consistent engine, with a full breakdown.

05 · REPORT
Results & exports
Filter, inspect any answer sheet, print, export to CSV.

> "We run midterms and finals on it — a whole class starts at the same time, and if the lab network drops, the students keep answering and the papers still arrive."
>
> — exam workflow in production at Kalar Technical Institute, Garmian Polytechnic University

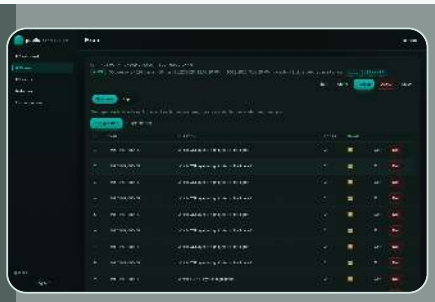
One system, every surface



Admin sign-in



Student start screen

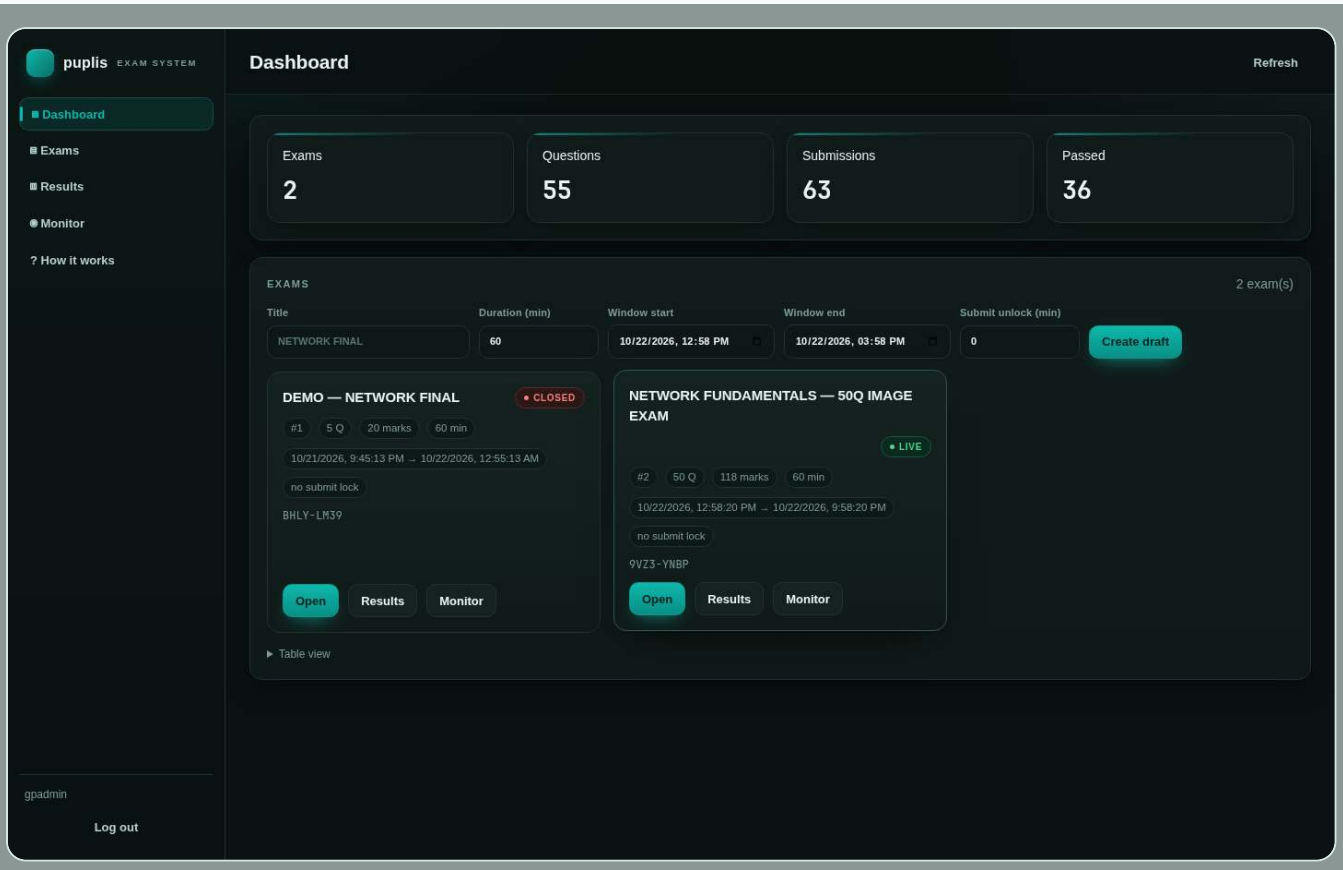


Exam & question management

FOR THE EXAM COMMITTEE

A serious admin, not a pile of forms.

A side panel and card grid put the four jobs where you expect them: **Dashboard**, **Exams**, **Results** and **Monitor**. Each page fits one screen — no hunting, no page scrollbars.



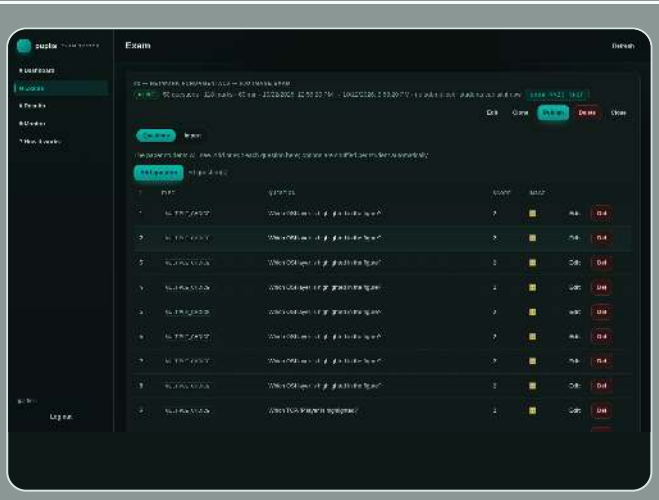
Dashboard: exams, questions, submissions and passes; every exam card carries state, marks, window and submit lock.

Build or import a paper in minutes

Six question types with dedicated editors — multiple choice, true/false, true/false with a correction blank, fill-in-the-blank, matching and multi-answer with points. Or import a paper:

- **8-column CSV** — the committee's established template, one row per question
- **JSON** for teams that generate papers, including the combined type
- **Dry run** first: every row is validated and reported before import

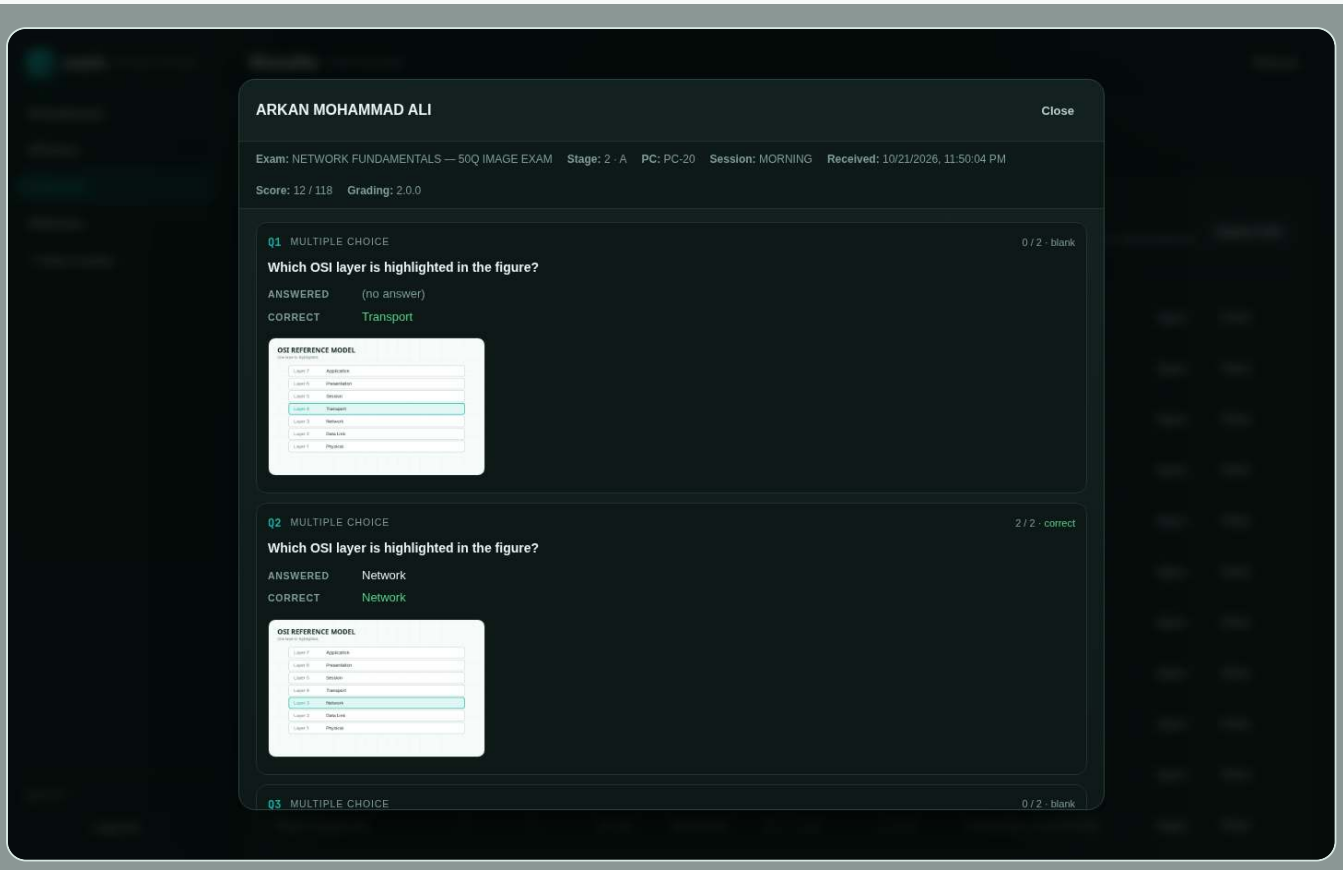
Publishing gives each exam a short **exam code** (for example `9VZ3-YNBP`) — share it and students are in.



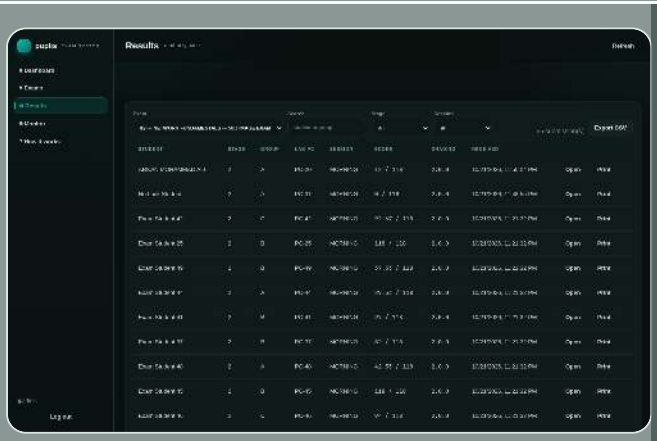
RESULTS YOU CAN DEFEND

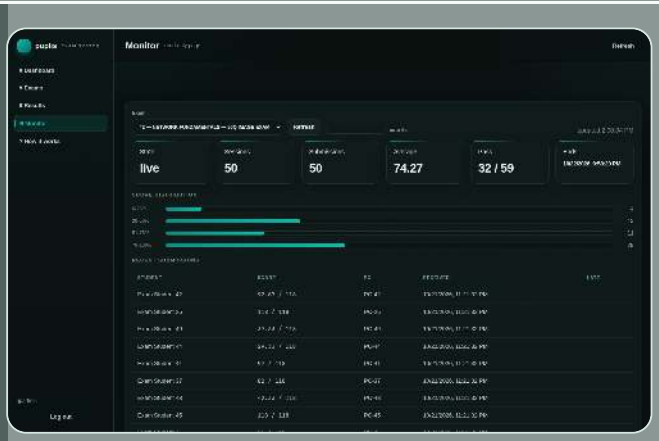
Open any student and see exactly how they were marked.

Results are filterable, exportable to CSV and printable. One click on **Open** reveals the entire answer sheet: what the student answered, the correct answer, the verdict and the marks for every question.



Student answers next to the key, per question, with marks and verdict — the answer to every mark dispute.





Results with filters and CSV export · live monitor with score distribution and recent submissions.

> Marking policy is explicit: partial credit where you choose it, negative marking with a documented floor, and a submit lock that is **optional per exam** — 0 minutes means students can submit immediately.

ALL THE FORMATS YOUR PAPERS ALREADY USE

Six question types. One grading engine.

Papers are rarely one format. puplis covers every type a technical institute uses — each with explicit marks, optional negative marking, and a per-question breakdown that explains every result.

01

Multiple choice

Four or more options, one or several correct. Options are shuffled per student, so a memorised letter from the next PC is wrong.

Marking: exact set · optional proportional credit

02

True / False

Fast to write, fast to answer. Ideal for checking definitions and common misconceptions.

Marking: exact flag, with optional negative score

03 NEW

True / False + correction

The student says whether the statement is true. If it is false, they must type the correction — so the question tests understanding, not just spotting a false statement.

Marking: full for flag + correction · half if only the flag is right

04

Fill in the blank

One or many blanks in a sentence or code fragment, with an explicit list of accepted wordings per blank.

Marking: partial credit per blank · accepted variants

05

Matching

Pair terms with definitions, protocols with purposes, connectors with cables. One clean dropdown per item.

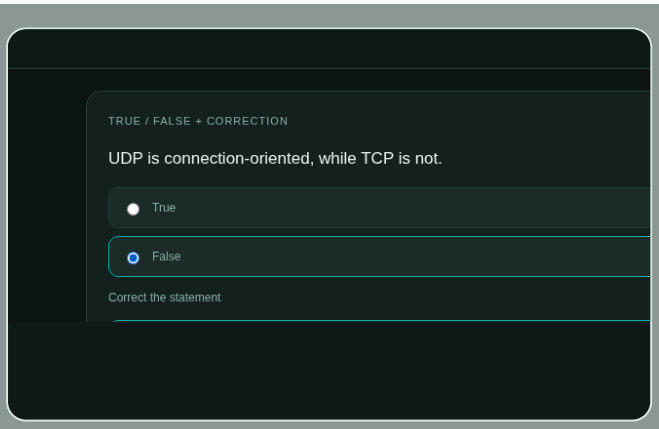
Marking: partial credit per correct pair

06

Multiple options with points

“Choose all that apply” with weighted options — partial selections earn their points, wrong ticks cost nothing unless you want them to.

Marking: points of selected options are summed



The answer sheet shows both: what the student chose, the correction they typed, and the accepted answers for every question.

New: true/false + correction

A false statement is no longer a coin flip. The student selects **False** and the correction blank opens; the answer is marked against the accepted wordings your lecturer supplied.

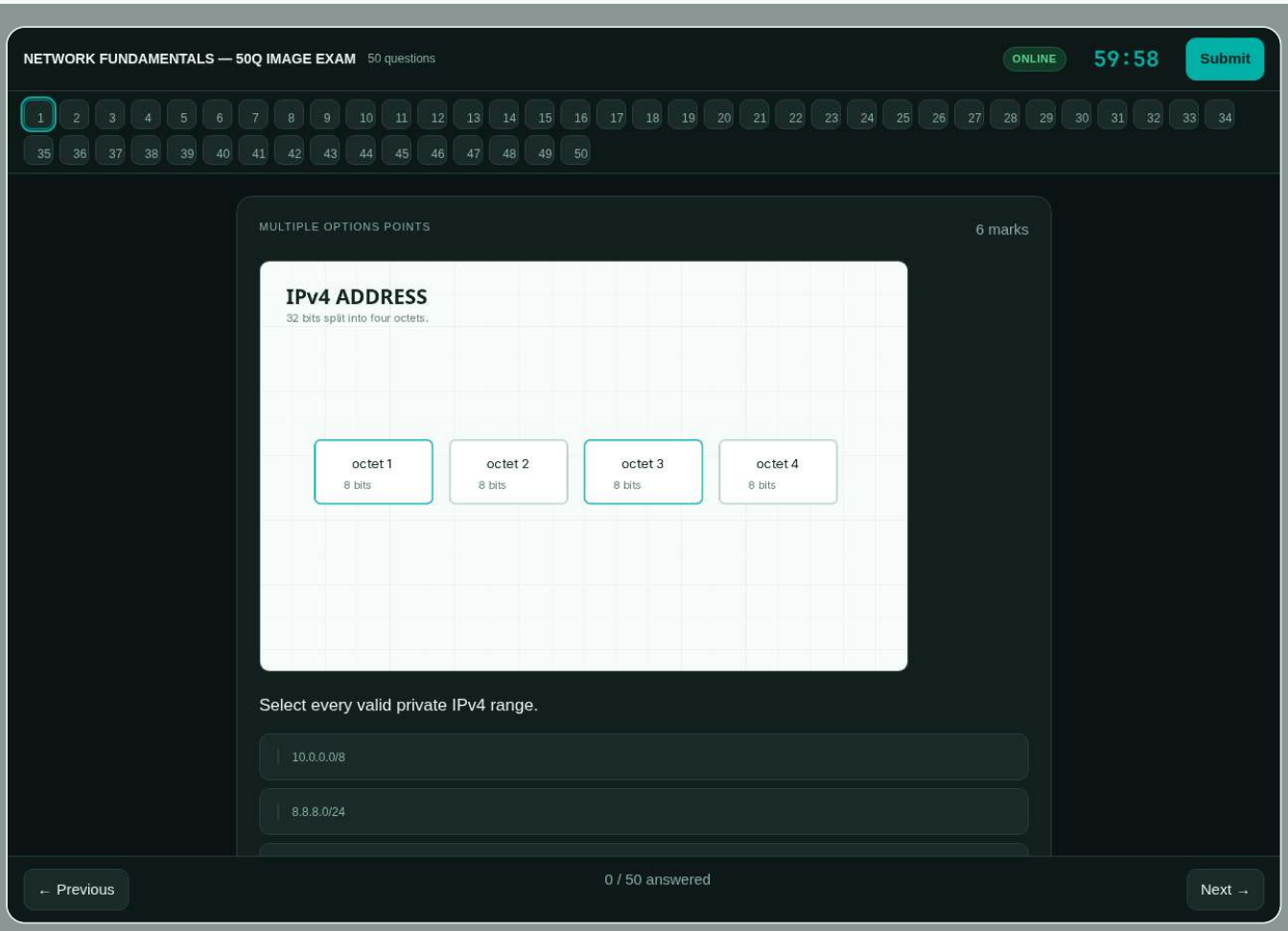
- Works in the admin editor, CSV and JSON import
- Multiple accepted corrections (e.g. *connectionless* / *connection less*)
- Mirrored in the answer sheet: *“False — correction: connectionless”*

> Every type above is already in daily use at Kalar Technical Institute — across 57 exams and more than 5,700 results.

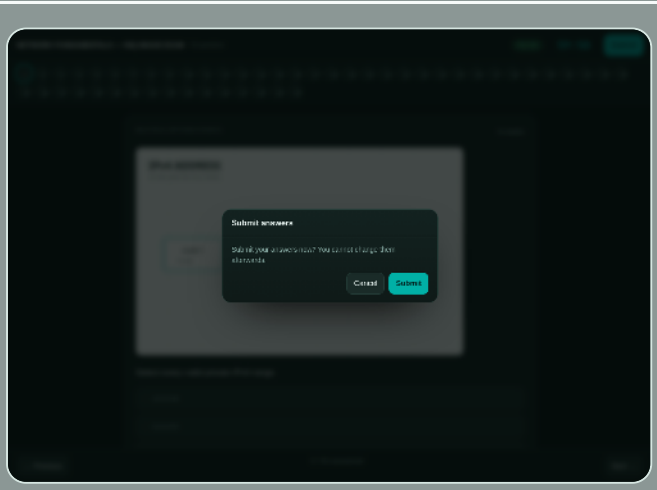
FOR THE STUDENT

One question per screen.
No app install. No numbers to cheat with.

Students open a URL, enter their name, stage, group and lab PC, and choose the live exam. The paper loads once, questions are shuffled per student, and the toolbar keeps the timer and submit button always in reach.



A real diagram question with shuffled options, numbered navigation boxes and a live timer — one screen, no scrolling.



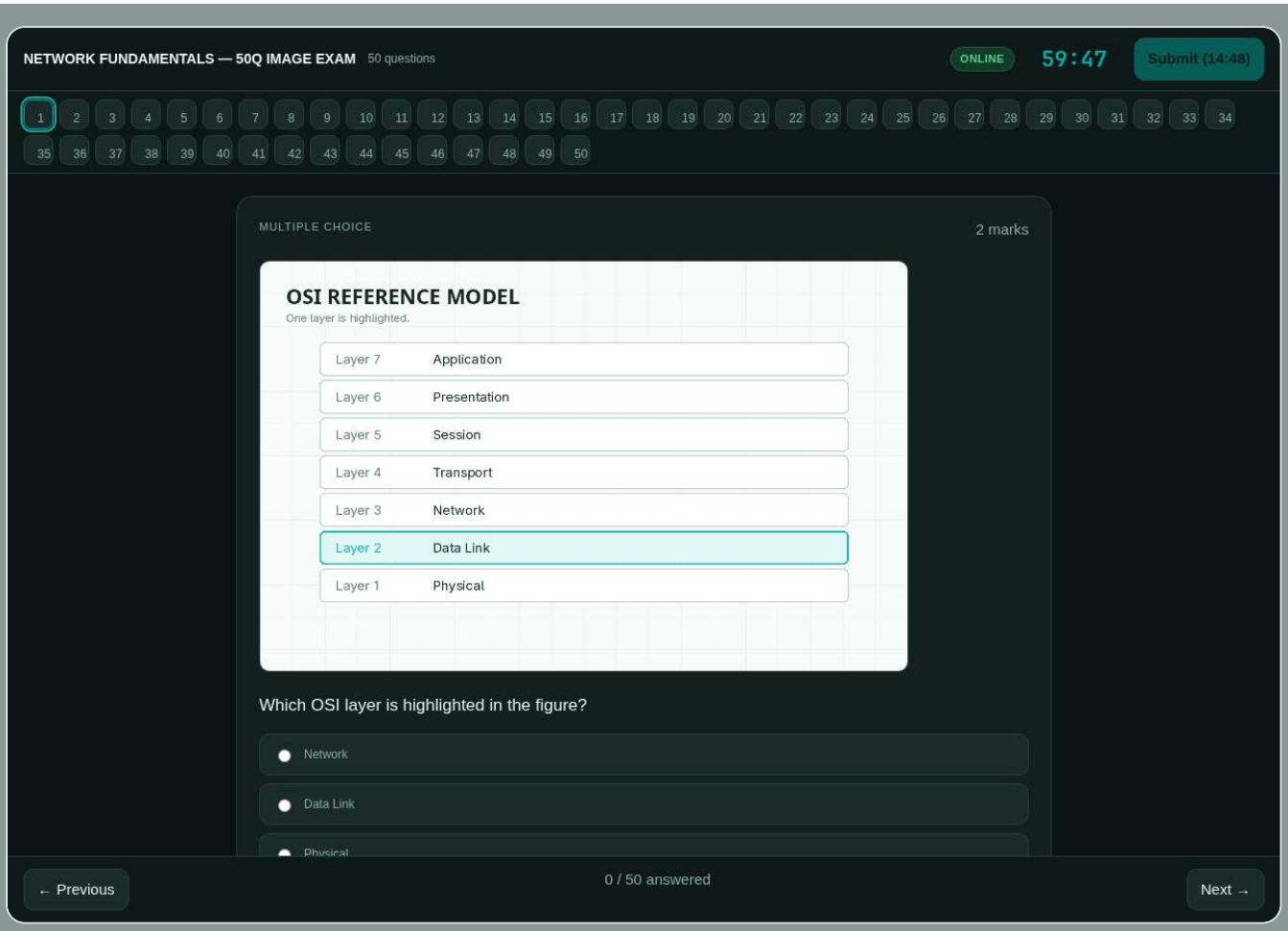
Built for lab reality

- **Question order shuffled per student**, stored so a refresh does not reshuffle.
- **Options shuffled** — a memorised letter from the next PC is wrong.
- **No question numbers** shown while answering; the boxes are progress only.
- **Custom dialogs** — no browser popups.
- **Optional submit lock** — set per exam, 0 minutes means submit anytime.
- **Resume** — a refresh brings back the same paper and answers.

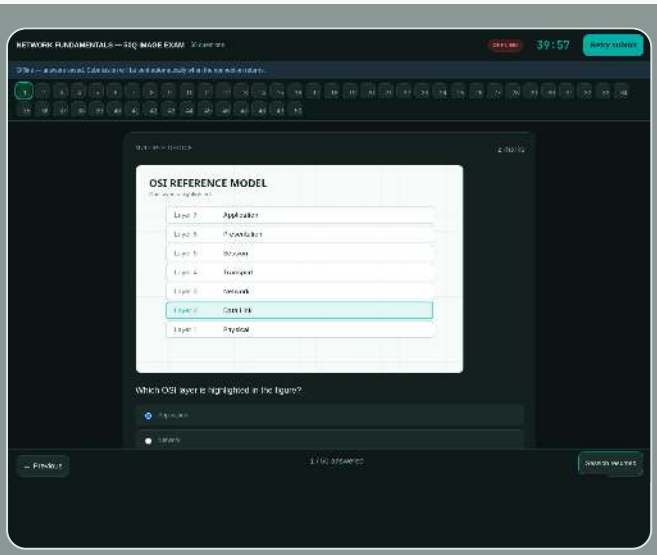
WHEN THE LINK GOES DOWN

A dropped cable is not a failed exam.

At start, the whole paper and every image are downloaded and cached on the student machine — questions in *localStorage*, images in *IndexedDB*, the app shell in a service worker. After that the sitting runs with **zero network**. Answers are saved on every keystroke; a submission made offline is queued and sent automatically when the connection returns, and it can never double-submit.



A 50-question image exam rendered with the network cut: all 50 boxes navigable, every image from cache.



Measured on a 50-question image exam

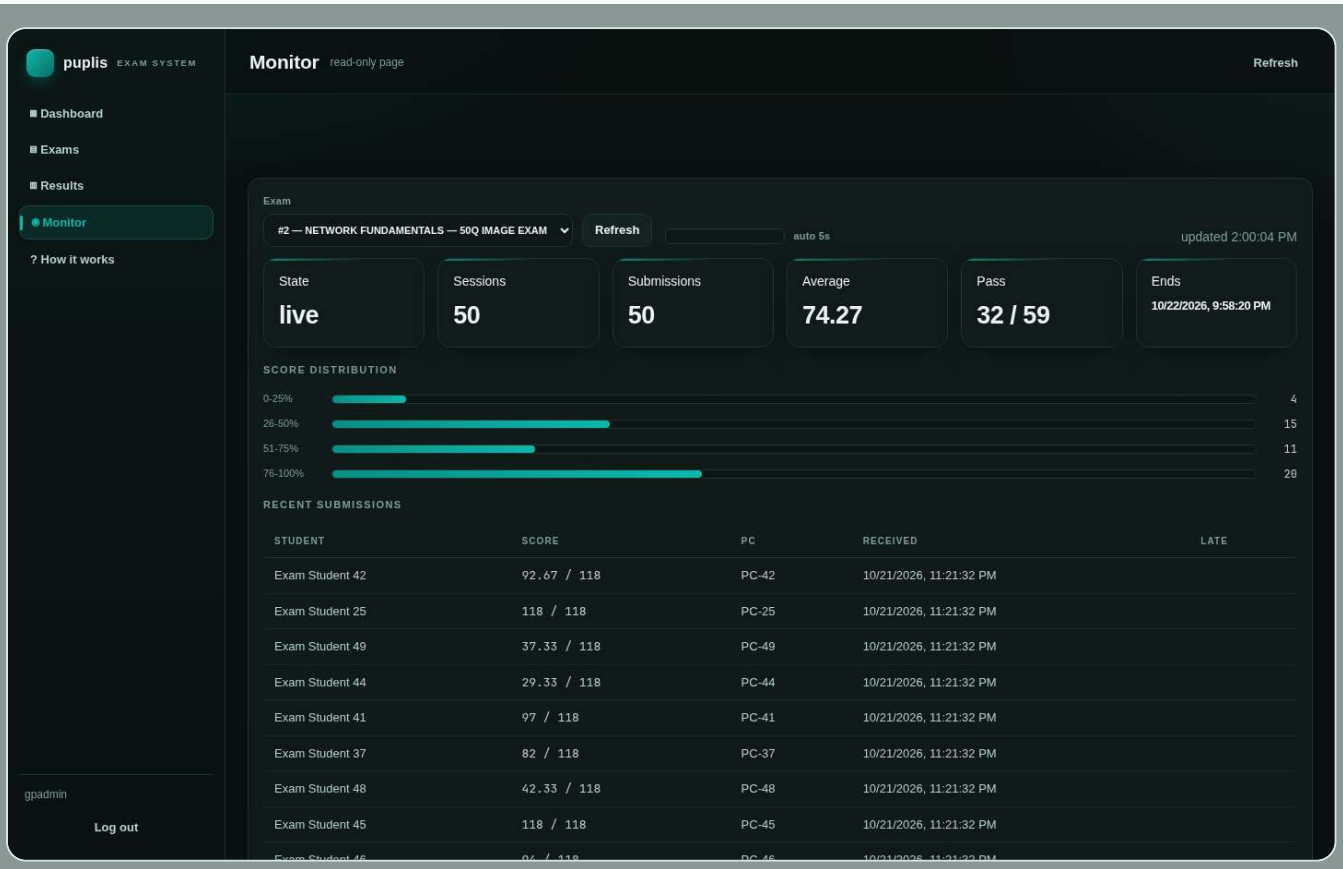
Start + prefetch	845 ms
Images cached	31 · 1.07 MB
Offline navigation	50 / 50
Image failures offline	0
Media requests offline	0
Reconnect → submit	automatic

PROOF UNDER LOAD

50 students at once is a non-event.

A full class sitting was simulated against the running system — 50 sessions started simultaneously, every student downloaded their images, answered and submitted.

648 ms	3.5 s	592 ms	74.3
50 starts, 0 errors	53 MB of images delivered	50 submissions, 0 errors	average / 118 marks



Live monitor after the burst: 50 submissions, average, pass count, score distribution and per-student marks.

Measured on laptop-class hardware. Scores ranged from 0 to a full 118/118, confirming the grading engine was marking every answer, not returning empty results.

DEPLOYMENT & OWNERSHIP

One lab PC. Two ports. Nothing in the cloud.

The system is self-hosted: an admin app on port 8013 for the committee and a student app on 8014 for the labs, sharing one database with WAL journaling and consistent hot backups. A service restarts it after a reboot; images are compressed and content-addressed; sessions and submissions are indexed for fast results.

Built to be trusted

- **Comprehensive automated test suite**, including grading and the full API flow
- **One consistent grading engine** — every result marked the same way
- **Security:** hashed session tokens, login rate limiting, payload limits, hardened headers, separate surfaces
- **No answers leak:** student responses never contain keys, points or scores
- **Data stays yours:** the database is a file on your own machine, with scheduled backups

Ready to grow

- SQLite today; **PostgreSQL-ready** for multi-institute use
- Exam management, question types and CSV/JSON import — all scriptable through the API
- Printable result pages and CSV exports for exam boards
- OpenAPI documentation at /docs on the admin port
- Runs on the hardware you already have — no cloud subscription

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puplis — exam system

In daily use at Kalar Technical Institute / Garmian Polytechnic University.

Design, backend and deployment by **Kisra Tech Solutions** — Kalar, Sulaymaniyah, Kurdistan Region, Iraq.

web admin · web student · offline · 50-student tested · self-hosted · consistency